SALES PULSE
Sales and Demand Forecasting System

Final Project Review | Department of CSE | KLH Deemed to be University

TEAM MEMBERS

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Guide: Dr . N. Sunanda - Associate Professor | Academic Year: 2026-27

Introduction & Problem Statement

Background

Businesses use historical sales data to understand product performance, sales trends and changes in customer demand. Analyzing this data helps businesses plan future sales and manage inventory more effectively.

Problem Statement

Traditional sales systems mainly record transactions and display past sales, but provide limited support for predicting future demand. This makes it difficult for businesses to identify demand changes and plan inventory in advance.

Motivation / Need

Sales Pulse aims to convert historical sales data into meaningful forecasts and visual insights. It supports business owners and sales teams in understanding demand patterns and making better-informed sales and inventory decisions.



Objectives & Scope

Objectives

1. **Develop and evaluate time-series forecasting models** and present historical sales, predicted demand and insights through a dashboard.

2. To analyze historical sales data and identify sales trends **over different time periods**.

3. **To compare sales revenue** across different product categories and regions.

4. **To calculate product-wise sales revenue**, units sold, and identify the best-selling and least-selling products.

Literature Survey

S.No	Author(s) & Year	Method / Technique	Key Findings	Limitation / Gap
1	Research paper authors (2022)	Machine Learning, Time-Series Forecasting	Studies forecasting of seasonal product sales.	Focuses mainly on furniture sales; may not generalize to other categories.
2	Hasan et al. (2022)	Retail Sales Forecasting	Compares forecasting approaches using Walmart sales data.	Performance may vary across businesses and datasets.
3	Research paper authors (2022)	ARIMA	Compares statistical and deep-learning approaches for retail demand forecasting.	Uses data from a specific Austrian retailer; results depend on demand patterns and promotions.
4	Dmitry Brykin (2024)	ARIMA, Prophet	Compares different forecasting models for sales prediction.	Model performance may vary across sales categories and forecasting periods.



Research Gap

Existing studies often focus on specific datasets, product categories or forecasting approaches, so their results may not generalize to different business datasets. Sales Pulse addresses this by providing an integrated workflow for CSV-based sales data, preprocessing, exploratory analysis, time-series forecasting, model evaluation and dashboard visualization to support sales and inventory planning.

Proposed System & Methodology

STEP 1

Define Problem

Identify the need to predict future sales using historical sales data..

STEP 2

Collect Data

Upload sales data through a CSV file containing date and sales-related columns.

STEP 3

Design

Design a forecasting workflow using Python, Pandas, ARIMA and Streamlit.

STEP 4

Implement

Develop CSV upload, data preprocessing, historical sales visualization and forecasting modules.

STEP 5

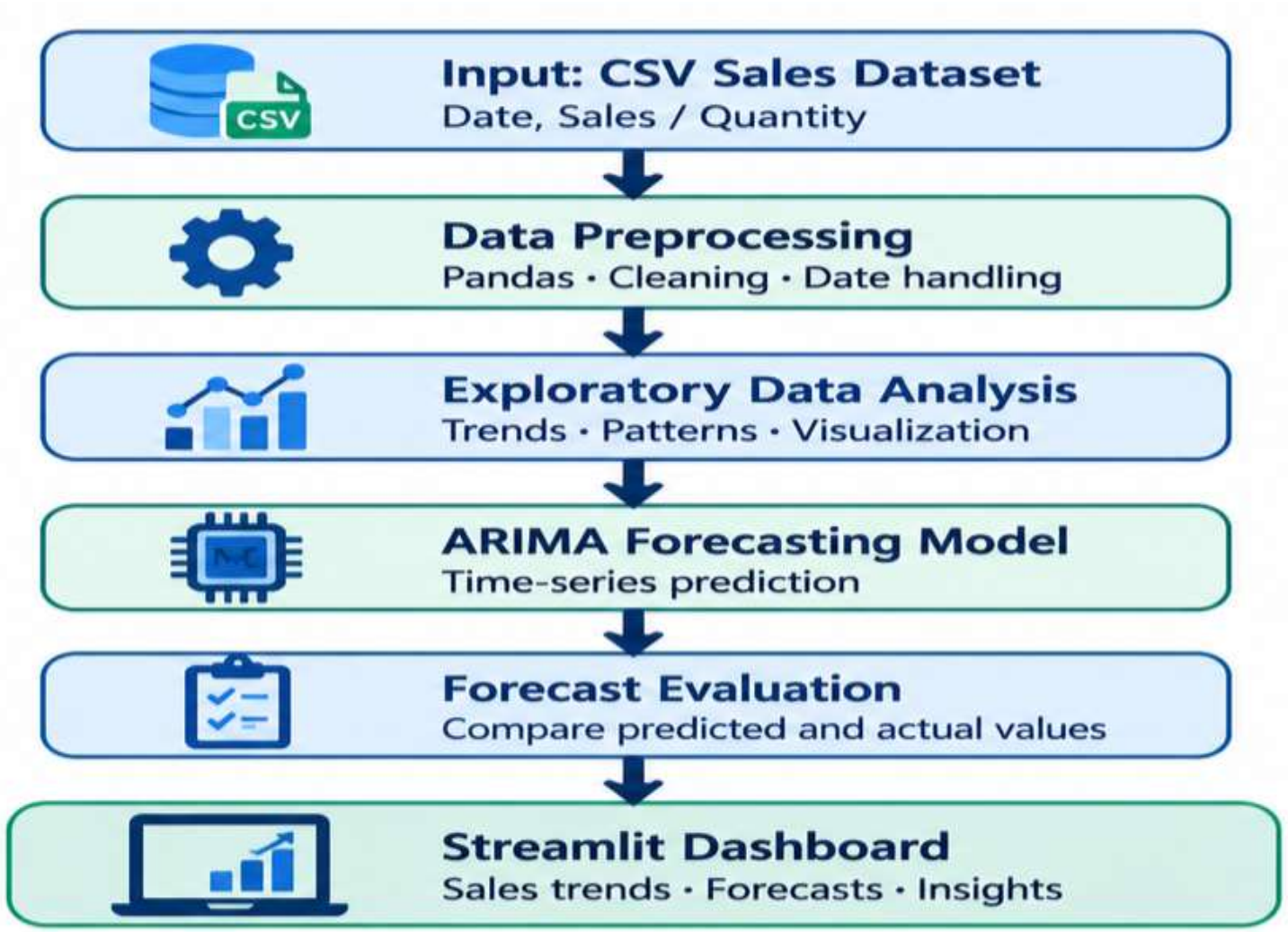
Test & Evaluate

Test uploaded datasets, generate forecasts and compare predictions with actual sales using evaluation metrics.

Key Novelty / Contribution

Sales Pulse provides a simple, user-friendly dashboard that accepts user-uploaded sales CSV files instead of depending on one fixed dataset. It combines data preprocessing, historical sales analysis and ARIMA-based forecasting in one application.

System Architecture



Components

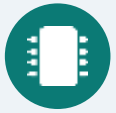
1. **Data Input**
   Uploads historical sales data through CSV files.

2. **Data Preprocessing**
   Cleans and prepares data for analysis.

3. **Forecasting Engine**
   Uses the ARIMA model to predict future sales.

4. **Dashboard and Evaluation**
   Displays sales trends, forecasts and evaluation results.

Hardware & Software Requirements



Hardware

- Laptop / Desktop Computer
- Multi-core Processor
- Minimum 4 GB RAM recommended
- Sufficient storage for datasets and project files



Software & Tools

- Windows 11
- Python 3.12
- Visual Studio Code (VS Code)
- Git and GitHub



Libraries / Frameworks

- Pandas – Data processing
- NumPy – Numerical operations
- Statsmodels – ARIMA forecasting
- Streamlit – Dashboard development

Deployment : Local machine(Port 8051)

Dataset / Data Understanding

[9,995]

Records (rows)

[21]

Features (columns)

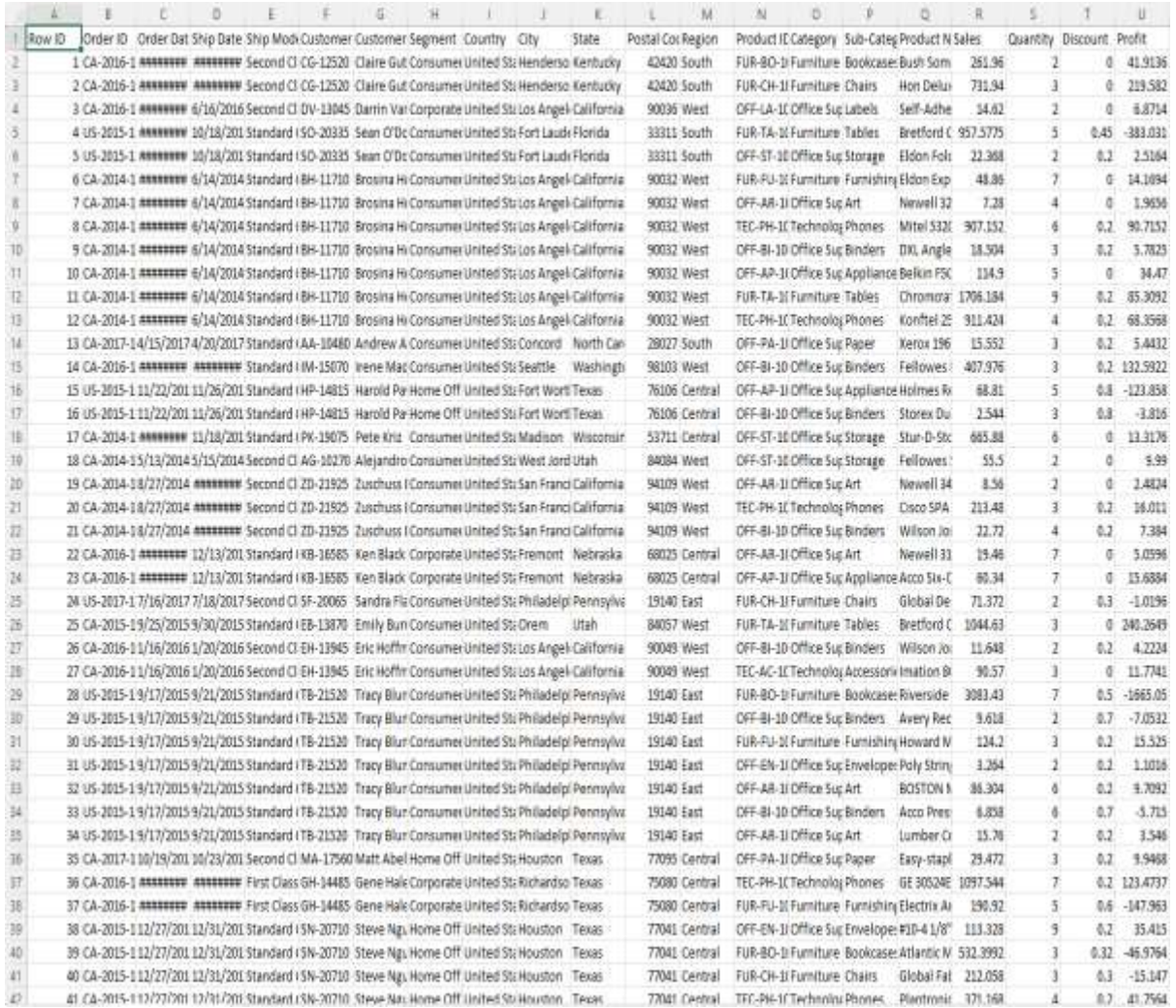
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Missing values

[NA]

Target classes

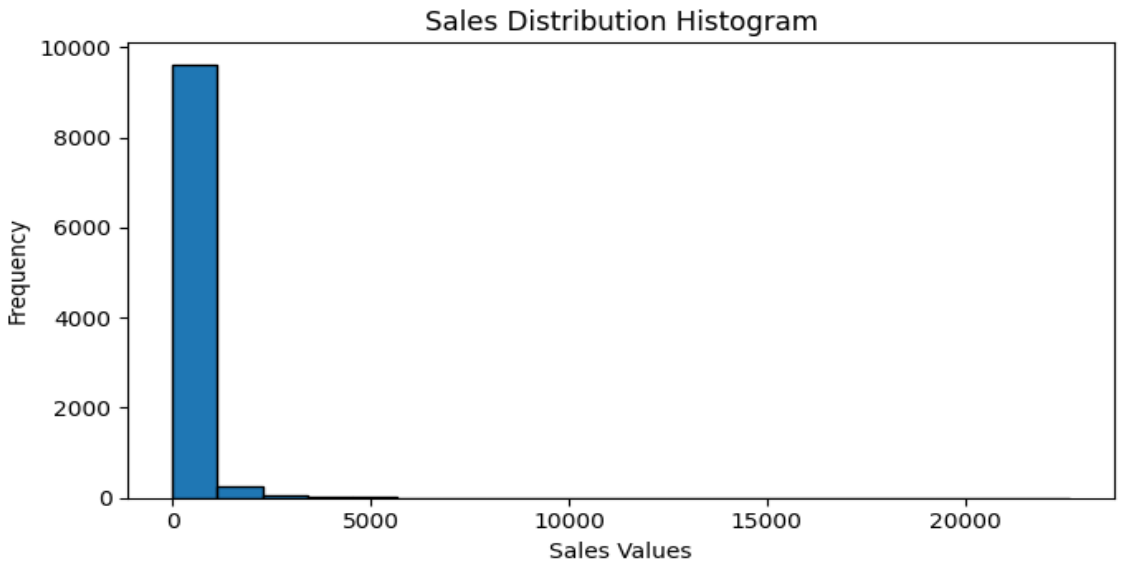
Superstore.csv

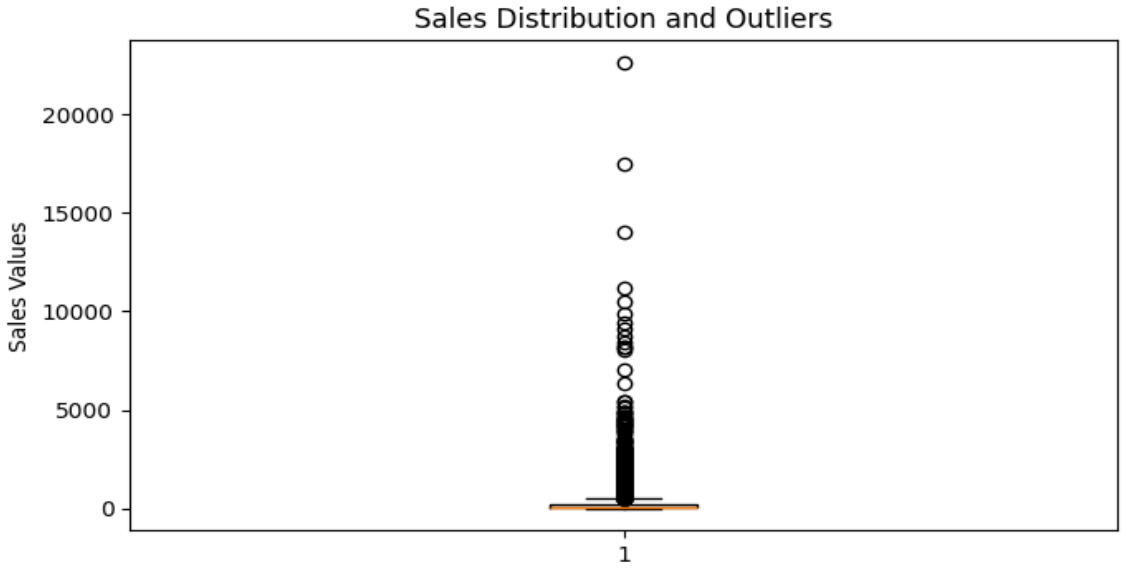


Row ID	Order ID	Order Dat	Ship Date	Ship Mod	Customer	Customer	Segment	Country	City	State	Postal Co	Region	Product ID	Category	Sub-Categ	Product N	Sales	Quantity	Discount	Profit
1	CA-2016-1	########	########	Second Cl	CG-12520	Claire Gut	Consumer	United Sta	Henderso	Kentucky	42420	South	FUR-BO-1	Furniture	Bookcase	Bush Som	261.96	2	0	41.9136
2	CA-2016-1	########	########	Second Cl	CG-12520	Claire Gut	Consumer	United Sta	Henderso	Kentucky	42420	South	FUR-CH-1	Furniture	Chairs	Hon Delu	731.94	3	0	219.582
3	CA-2016-1	########	6/16/2016	Second Cl	DV-13045	Darrin Van	Corporate	United Sta	Los Angel	California	90036	West	OFF-LA-1	Office Sup	Labels	Self-Adhe	14.62	2	0	6.8714
4	US-2015-1	########	10/18/201	Standard	SO-20335	Sean O'Do	Consumer	United Sta	Fort Laude	Florida	33311	South	FUR-TA-1	Furniture	Tables	Bretford C	957.5775	5	0.45	-383.031
5	US-2015-1	########	10/18/201	Standard	SO-20335	Sean O'Do	Consumer	United Sta	Fort Laude	Florida	33311	South	OFF-ST-1	Office Sup	Storage	Eldon Fol	22.368	2	0.2	2.5164
6	CA-2014-1	########	6/14/2014	Standard	BH-11710	Brosina Ho	Consumer	United Sta	Los Angel	California	90032	West	FUR-FU-1	Furniture	Furnishing	Eldon Exp	48.86	7	0	14.1694
7	CA-2014-1	########	6/14/2014	Standard	BH-11710	Brosina Ho	Consumer	United Sta	Los Angel	California	90032	West	OFF-AR-1	Office Sup	Art	Newell 32	7.28	4	0	1.9656
8	CA-2014-1	########	6/14/2014	Standard	BH-11710	Brosina Ho	Consumer	United Sta	Los Angel	California	90032	West	TEC-PH-1	Technolog	Phones	Mitel 532	907.152	6	0.2	90.7152
9	CA-2014-1	########	6/14/2014	Standard	BH-11710	Brosina Ho	Consumer	United Sta	Los Angel	California	90032	West	OFF-BI-10	Office Sup	Binders	DXL Angle	18.504	3	0.2	5.7825
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11	CA-2014-1	########	6/14/2014	Standard	BH-11710	Brosina Ho	Consumer	United Sta	Los Angel	California	90032	West	FUR-TA-1	Furniture	Tables	Chromcra	1706.184	9	0.2	85.3092
12	CA-2014-1	########	6/14/2014	Standard	BH-11710	Brosina Ho	Consumer	United Sta	Los Angel	California	90032	West	TEC-PH-1	Technolog	Phones	Konftel 2	911.424	4	0.2	68.3568
13	CA-2017-1	4/15/2017	4/20/2017	Standard	AA-10480	Andrew A	Consumer	United Sta	Concord	North Car	28027	South	OFF-PA-1	Office Sup	Paper	Xerox 196	15.552	3	0.2	5.4432
14	CA-2016-1	########	########	Standard	IM-15070	Irene Mac	Consumer	United Sta	Seattle	Washingt	98103	West	OFF-BI-10	Office Sup	Binders	Fellowes	407.976	3	0.2	132.5922
15	US-2015-1	11/22/201	11/26/201	Standard	HP-14815	Harold Pa	Home Off	United Sta	Fort Wort	Texas	76106	Central	OFF-AP-1	Office Sup	Appliance	Holmes Re	68.81	5	0.8	-123.858
16	US-2015-1	11/22/201	11/26/201	Standard	HP-14815	Harold Pa	Home Off	United Sta	Fort Wort	Texas	76106	Central	OFF-BI-10	Office Sup	Binders	Storex Du	2.544	3	0.8	-3.816
17	CA-2014-1	########	11/18/201	Standard	PK-19075	Pete Kriz	Consumer	United Sta	Madison	Wisconsin	53711	Central	OFF-ST-1	Office Sup	Storage	Stur-D-Sto	665.88	6	0	13.3176
18	CA-2014-1	5/13/2014	5/15/2014	Second Cl	AG-10270	Alejandro	Consumer	United Sta	West Jord	Utah	84084	West	OFF-ST-1	Office Sup	Storage	Fellowes	55.5	2	0	9.99
19	CA-2014-1	8/27/2014	########	Second Cl	ZD-21925	Zuschuss I	Consumer	United Sta	San Franci	California	94109	West	OFF-AR-1	Office Sup	Art	Newell 34	8.56	2	0	2.4824
20	CA-2014-1	8/27/2014	########	Second Cl	ZD-21925	Zuschuss I	Consumer	United Sta	San Franci	California	94109	West	TEC-PH-1	Technolog	Phones	Cisco SPA	213.48	3	0.2	16.011
21	CA-2014-1	8/27/2014	########	Second Cl	ZD-21925	Zuschuss I	Consumer	United Sta	San Franci	California	94109	West	OFF-BI-10	Office Sup	Binders	Wilson Jo	22.72	4	0.2	7.384
22	CA-2016-1	########	12/13/201	Standard	KB-16585	Ken Black	Corporate	United Sta	Fremont	Nebraska	68025	Central	OFF-AR-1	Office Sup	Art	Newell 33	19.46	7	0	5.0596
23	CA-2016-1	########	12/13/201	Standard	KB-16585	Ken Black	Corporate	United Sta	Fremont	Nebraska	68025	Central	OFF-AP-1	Office Sup	Appliance	Acco Six-C	60.34	7	0	15.6884
24	US-2017-1	7/16/2017	7/18/2017	Second Cl	SF-20065	Sandra Fla	Consumer	United Sta	Philadelpi	Pennsylva	19140	East	FUR-CH-1	Furniture	Chairs	Global De	71.372	2	0.3	-1.0196
25	CA-2015-1	9/25/2015	9/30/2015	Standard	EB-13870	Emily Bun	Consumer	United Sta	Orem	Utah	84057	West	FUR-TA-1	Furniture	Tables	Bretford C	1044.63	3	0	240.2649
26	CA-2016-1	1/16/2016	1/20/2016	Second Cl	EH-13945	Eric Hoffm	Consumer	United Sta	Los Angel	California	90049	West	OFF-BI-10	Office Sup	Binders	Wilson Jo	11.648	2	0.2	4.2224
27	CA-2016-1	1/16/2016	1/20/2016	Second Cl	EH-13945	Eric Hoffm	Consumer	United Sta	Los Angel	California	90049	West	TEC-AC-1	Technolog	Accessorie	Imation 8	90.57	3	0	11.7741
28	US-2015-1	9/17/2015	9/21/2015	Standard	TB-21520	Tracy Blur	Consumer	United Sta	Philadelpi	Pennsylva	19140	East	FUR-BO-1	Furniture	Bookcase	Riverside	3083.43	7	0.5	-1665.05
29	US-2015-1	9/17/2015	9/21/2015	Standard	TB-21520	Tracy Blur	Consumer	United Sta	Philadelpi	Pennsylva	19140	East	OFF-BI-10	Office Sup	Binders	Avery Rec	9.618	2	0.7	-7.0532
30	US-2015-1	9/17/2015	9/21/2015	Standard	TB-21520	Tracy Blur	Consumer	United Sta	Philadelpi	Pennsylva	19140	East	FUR-FU-1	Furniture	Furnishing	Howard M	124.2	3	0.2	15.525
31	US-2015-1	9/17/2015	9/21/2015	Standard	TB-21520	Tracy Blur	Consumer	United Sta	Philadelpi	Pennsylva	19140	East	OFF-EN-1	Office Sup	Envelopes	Poly Strin	3.264	2	0.2	1.1016
32	US-2015-1	9/17/2015	9/21/2015	Standard	TB-21520	Tracy Blur	Consumer	United Sta	Philadelpi	Pennsylva	19140	East	OFF-AR-1	Office Sup	Art	BOSTON M	86.304	6	0.2	9.7092
33	US-2015-1	9/17/2015	9/21/2015	Standard	TB-21520	Tracy Blur	Consumer	United Sta	Philadelpi	Pennsylva	19140	East	OFF-BI-10	Office Sup	Binders	Acco Pres	6.858	6	0.7	-5.715
34	US-2015-1	9/17/2015	9/21/2015	Standard	TB-21520	Tracy Blur	Consumer	United Sta	Philadelpi	Pennsylva	19140	East	OFF-AR-1	Office Sup	Art	Lumber Cr	15.76	2	0.2	3.546
35	CA-2017-1	10/19/201	10/23/201	Second Cl	MA-17560	Matt Abel	Home Off	United Sta	Houston	Texas	77095	Central	OFF-PA-1	Office Sup	Paper	Easy-stapl	29.472	3	0.2	9.9468
36	CA-2016-1	########	########	First Class	GH-14485	Gene Hale	Corporate	United Sta	Richardso	Texas	75080	Central	TEC-PH-1	Technolog	Phones	GE 30524E	1097.544	7	0.2	123.4737
37	CA-2016-1	########	########	First Class	GH-14485	Gene Hale	Corporate	United Sta	Richardso	Texas	75080	Central	FUR-FU-1	Furniture	Furnishing	Electrix Ar	190.92	5	0.6	-147.963
38	CA-2015-1	12/27/201	12/31/201	Standard	SN-20710	Steve Nguy	Home Off	United Sta	Houston	Texas	77041	Central	OFF-EN-1	Office Sup	Envelopes	#10-4 1/8"	113.328	9	0.2	35.415
39	CA-2015-1	12/27/201	12/31/201	Standard	SN-20710	Steve Nguy	Home Off	United Sta	Houston	Texas	77041	Central	FUR-BO-1	Furniture	Bookcase	Atlantic M	532.3992	3	0.32	-46.9764
40	CA-2015-1	12/27/201	12/31/201	Standard	SN-20710	Steve Nguy	Home Off	United Sta	Houston	Texas	77041	Central	FUR-CH-1	Furniture	Chairs	Global Fal	212.058	3	0.3	-15.147
41	CA-2015-1	12/27/201	12/31/201	Standard	SN-20710	Steve Nguy	Home Off	United Sta	Houston	Texas	77041	Central	TEC-PH-1	Technolog	Phones	Plantronic	171.168	4	0.2	41.7564

- Source: User-uploaded sales dataset in CSV format.
- Features: Date and sales-related numerical columns.
- Quality: Dataset is checked for missing values, duplicate records and inconsistent formats.

Data Preprocessing & Exploratory Analysis





Cleaning

- Check missing values and duplicates.
- Handle inconsistent records.
- Remove invalid or unusable entries where necessary.

Transformation

- Convert date column into a consistent date format.
- Arrange records chronologically.
- Prepare sales values for time-series forecasting.

Feature Selection

- Select the date column as the time index.
- Select the sales column as the forecasting target.
- Use relevant numerical sales information.

Module-wise Description

1 Data Upload Module

Purpose: Allows users to upload sales datasets in CSV format.

Input → Output: CSV file → Uploaded dataset.

Developed by: None

2 Data Preprocessing Module

Purpose: Cleans and prepares sales data for forecasting.

Input → Output: Raw dataset → Processed time-series data.

Developed by: Nimisha

3 Sales Forecasting Module

Purpose: Predicts future sales and demand using historical sales data and time-series forecasting models such as ARIMA and Prophet.

Input → Output: Processed sales data → Forecast predictions and model comparison results.

Developed by: Sanuhya

4 Dashboard & Visualization Module

Purpose: Displays historical sales, forecast results and visual insights.

Input → Output: Forecast results → Interactive dashboard.

Developed by: Sarvajith

Algorithms & Technologies Used

ARIMA

Why chosen
A statistical time-series forecasting method used to predict future sales from historical sales patterns.

Key parameters

Autoregressive order (p), differencing order (d), moving-average order (q).

Prophet

Why chosen
Supports time-series forecasting by modelling trends and seasonal patterns in sales data.

Key parameters

Trend, seasonality, historical sales values, forecast horizon.

EDA

Why chosen
Helps identify sales trends, demand variations, and possible seasonal patterns before forecasting.

Key parameters

Sales over time, date-wise patterns, high- and low-demand periods.

Selection criteria: Forecast accuracy , suitability for the dataset, computational simplicity, and the ability to identify trends and seasonal patterns.

Implementation / Model Building

Feature Selection

Historical sales data using Date and Sales (Revenue) or Quantity as forecasting targets. Product, Category and Region are used for additional sales analysis.

Preprocessing Pipeline

Upload CSV/Excel files. → Identify and map relevant columns. → Handle invalid dates, missing sales values and duplicate records. → Convert sales data into daily totals and sort chronologically.

Data Splitting

Earlier historical observations are used for model training, while the latest 7–30 observations are reserved for testing.
Chronological splitting prevents future data leakage.

Configuration

Forecasting models: ARIMA and Prophet.
Input: Prepared historical sales time series.
Output: Predicted sales values for a future period.
Evaluation: Suitable forecasting error metrics.

Code snippet / pseudocode

```
# Prepare historical sales data
series = make_daily_series(data)

# Split data chronologically
training, testing = chronological_split(series)

# Compare forecasting models
models = ["Baseline", "ARIMA", "Prophet"]

for name in models:
    predictions = predict_with_model(
        name, training, testing["date"]
    )

    metrics = calculate_metrics(
        testing["target"], predictions
    )

# Select best model based on MAE
best_model = successful.sort_values("MAE").iloc[0]["Model"]

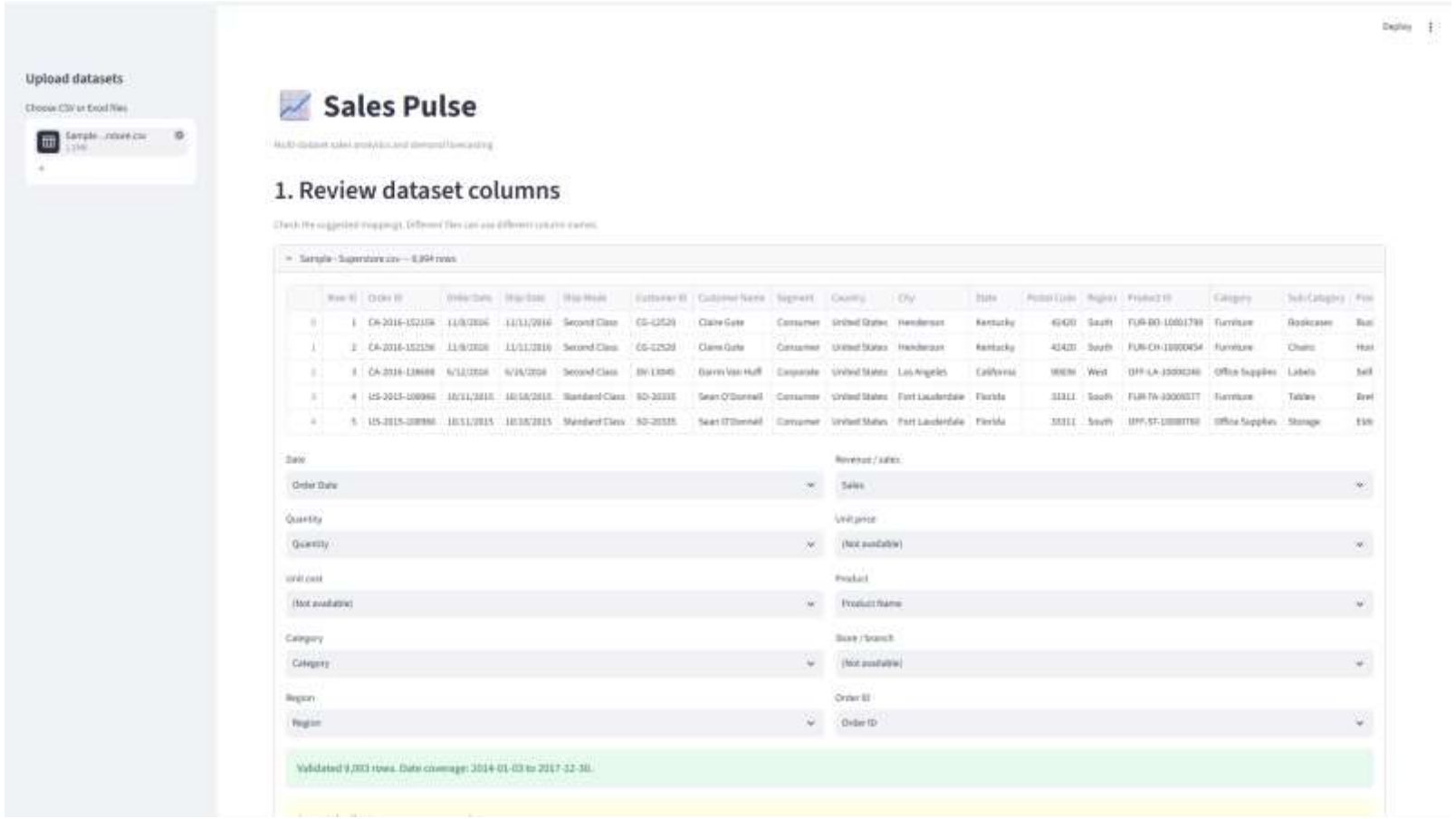
# Generate future sales predictions
forecast = predict_with_model(
    best_model, series, future_dates
)
```

Working Prototype / Output Screens



Output 1 : Dataset Upload
Upload CSV or Excel sales datasets for analysis.

Working Prototype / Output Screens



Output 2 : Dataset Validation

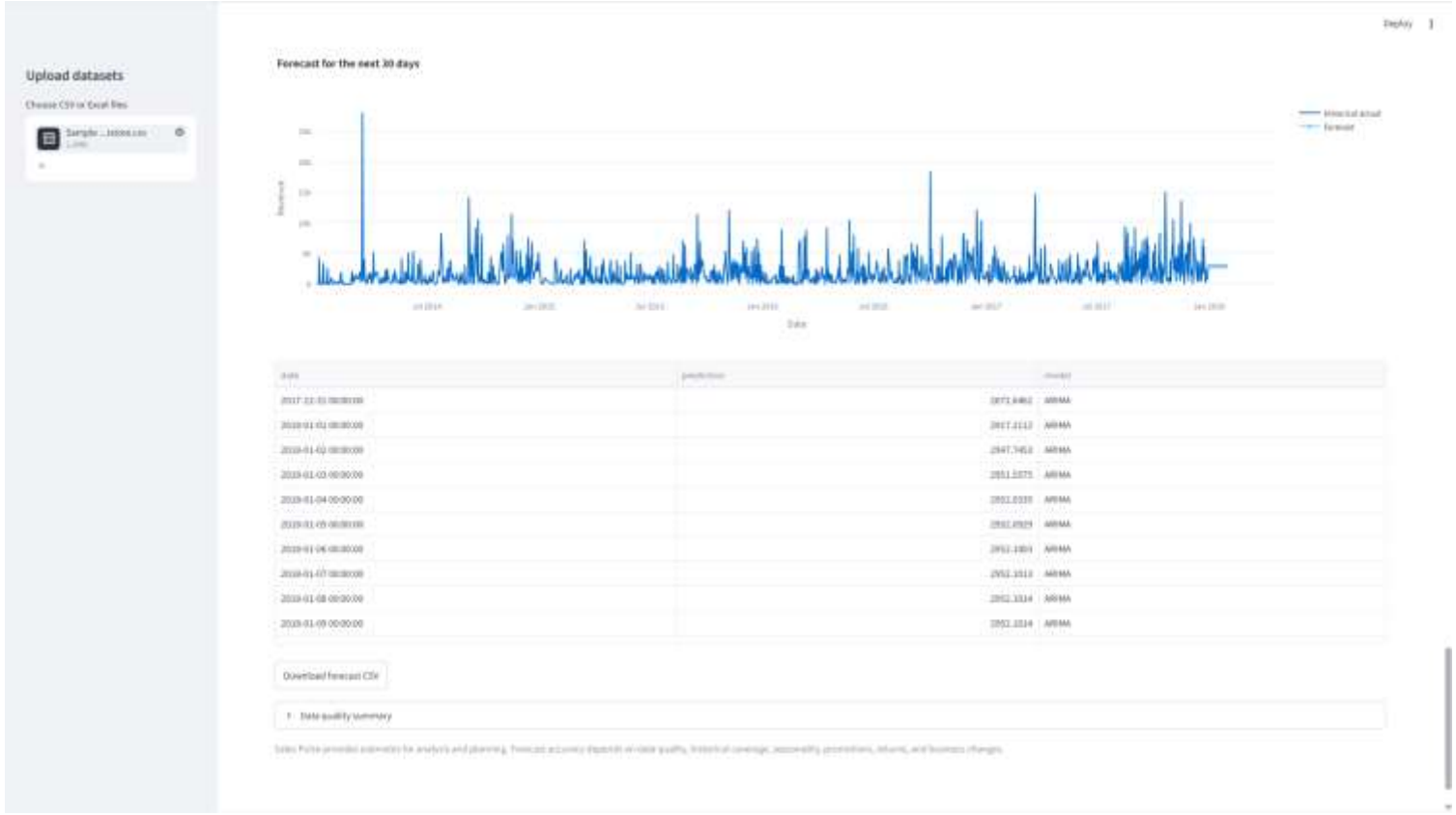
Displays dataset columns, maps sales fields, and validates the uploaded data.

Working Prototype / Output Screens



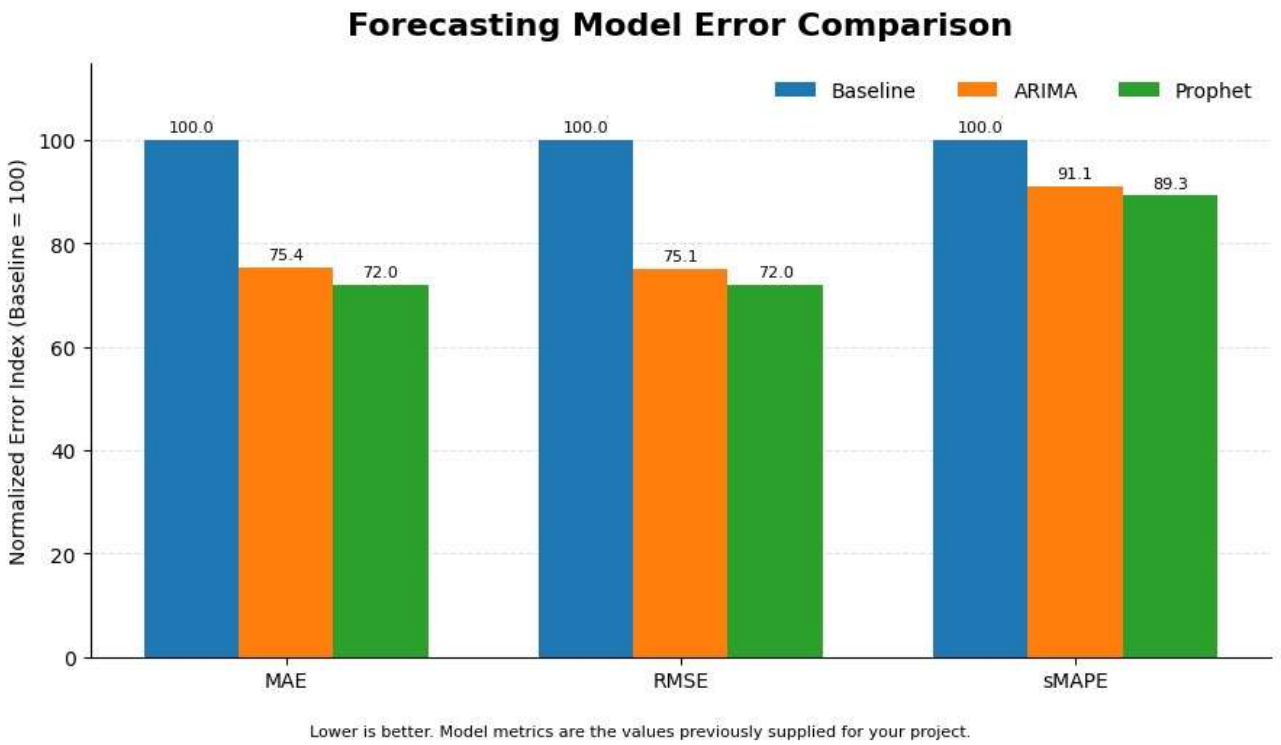
Output 3 : Sales Analytics Dashboard
Displays dataset columns, maps sales fields, and validates the uploaded data.

Working Prototype / Output Screens



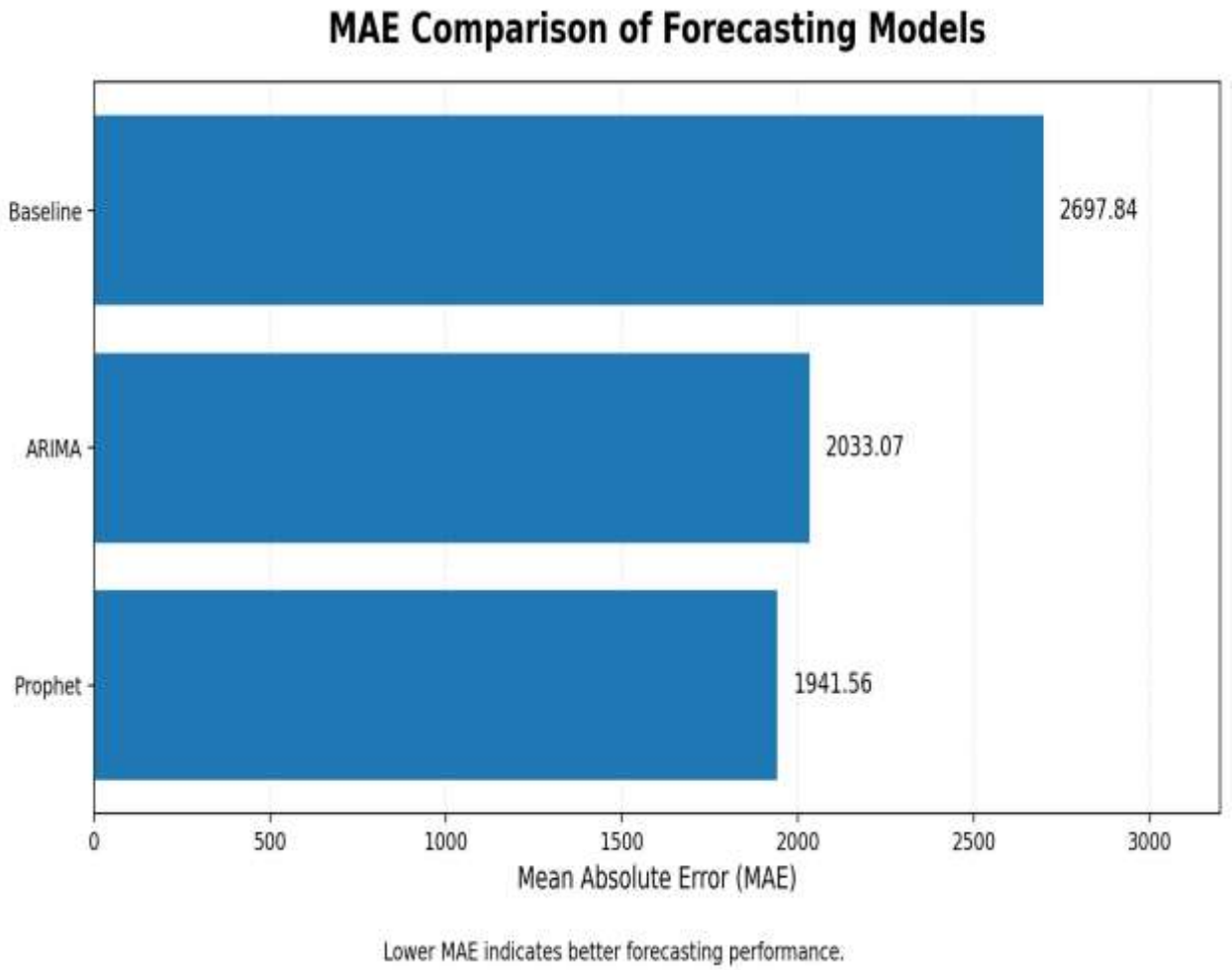
Output 4 : Run forecast
Displays future sales and demand

Results & Performance Comparison



Metric	Baseline	ARIMA	Prophet
MAE	2697.84	2033.07	1941.56
RMSE	3270.90	2457.03	2353.53
sMAPE	81.69%	74.39%	72.94%
Model Ranking	3rd	2nd	1st
Performance	Low	Good	Best

Analysis & Key Findings



Prophet

Best Performing Model
72.9388%

Interpretation: Prophet performed best among the three models by producing the lowest prediction error.
Error analysis: Forecast errors occur due to fluctuations in daily sales and changes in demand patterns.
Key drivers: Historical sales trends, daily demand variations and weekly seasonality.

Testing & Validation

Test ID	Test Description	Expected Result	Actual Result	Status
TC-01	Upload valid Superstore.csv file	Dataset should upload successfully	Dataset loaded for analysis	Pass
TC-02	Check dataset columns and data	Required date and sales columns should be identified	Dataset columns are checked	Pass
TC-03	Preprocess sales data	Data should be cleaned and arranged chronologically	Processed sales data is prepared	Pass
TC-04	Generate sales forecast	System should display predicted future sales	Forecast output is generated	Pass



Functional Testing

Tested CSV upload, data preprocessing, sales analysis and forecast generation.



Validation Method

- Historical sales data is divided chronologically into training and testing data.
- Forecasted values are compared with actual sales values.



Performance Testing

- Checked dataset loading and processing.
- Verified that forecasting results and dashboard visualizations are generated correctly.

Challenges & Limitations



Challenges Faced

- Data Cleaning: Handling missing values and inconsistent date formats → Applied data preprocessing.
- Time-Series Preparation: Converting sales records into chronological data → Sorted and aggregated sales data by date.
- Forecasting: Selecting a suitable forecasting approach → Used time-series forecasting based on historical sales patterns.



Limitations

- Forecasting performance depends on the quality and availability of historical Superstore data.
- External factors such as discounts, promotions and competitor activities are not considered.
- Predictions may be less reliable when demand patterns change unexpectedly.

Conclusion

1. Developed Sales Pulse, a sales and demand forecasting system that analyzes the Superstore.csv dataset using data preprocessing, exploratory analysis and time-series forecasting.

2. The system analyzes historical sales trends and generates future sales predictions, helping users understand possible changes in demand.

3. Integrated data analysis, forecasting and visualization into one application to support better sales planning and inventory management.



Final Outcome

Historical Sales Analysis & Future Demand Prediction

Future Scope & Applications

Improve Performance
- Integrate advanced forecasting models such as LSTM.
- Improve prediction accuracy through model tuning and comparison.

Expand Data / Features
- Support larger datasets and multiple product categories.
- Include external factors such as promotions and seasonal demand.

Deploy & Build an Interface
- Develop an interactive web-based dashboard.
- Add real-time sales updates and customizable forecast periods.

Scale & Secure
- Enable cloud-based deployment.
- Improve data security, reliability and scalability.

Applications
- Inventory and stock management.
- Sales and revenue planning.
- Retail and e-commerce demand forecasting.
- Business owners, retailers and sales analysts.

Team Contribution

Reg. No.	Name	Contribution / Modules Handled
2420030427	Y. Nimisha	Implementation
2420030498	K. Sanuhya	Project Report
2420030516	R. Sarvajith varma	PPT and Research paper

References

[1] “Time-series forecasting of seasonal items sales using machine learning – A comparative analysis,” Journal of Innovation & Knowledge, 2022. doi: 10.1016/j.jjimei.2022.100058.

[2] M. R. Hasan, M. A. Kabir, R. A. Shuvro, and P. Das, “A comparative study on forecasting of retail sales,” arXiv:2203.06848, 2022.

[3] “Predictive analytics for demand forecasting – A comparison of SARIMA and LSTM in retail SCM,” Procedia Computer Science, 2022. doi: 10.1016/j.procs.2022.01.298.

[4] D. Brykin, “Sales forecasting models: Comparison between ARIMA, LSTM and Prophet,” Journal of Computer Science, 2024. doi: 10.3844/jcssp.2024.1222.1230.

[5] H. Song, D. Du, and Y. Li, “Research on precise demand forecast of retail commodities based on LSTM-ARIMA joint model,” Journal of Physics: Conference Series, vol. 1865, no. 4, 2021.

[6] A. Ecevit, İ. Öztürk, M. Dağ, and T. Özcan, “Short-term sales forecasting using LSTM and Prophet based models in e-commerce,” 2023.

Add this dataset reference as [7]:

[7] Superstore.csv, historical retail sales dataset used for sales analysis and demand forecasting.



Project Links

GitHub: [URL]
Dataset: [URL]

THANK YOU

Questions & Discussion

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